



SKILL DEVELOPMENT CERTIFICATION CUM INTERNSHIP
Data Science Bootcamp: From Foundations to Real-World Projects
Indian Institute of Information Technology Kottayam
Department of Computer Science & Engineering

SYLLABUS

01 Deep Learning & Computer Vision

- Neural networks, MLPs & regularization
- CNNs: convolution, pooling & feature maps
- Transfer learning: ResNet & VGG
- Image classification pipelines & evaluation
- Explainable AI for computer vision
- Visualization and interpretation of model decision

02 Data Analytics for Healthcare

- Clinical Predictions using ANN
- Clinical Predictions using Decision trees
- Advanced prediction models with multiple instance learning and kernel methods
- Survival models with non parametric survival analysis
- Evaluation and validation for clinical predictions
- Visual analytics for patient data

03 Introduction to Network Science

- Understanding data as networks
- Centrality measures and network models
- Social network analysis
- Network robustness
- Graph based learning
- Tools: NetworkX, Gephi, CytoScape, VOSviewer

04 Praxis — Project-Based Learning

- Topic selection & scope definition
- Data exploration & preparation
- Design & implement analytical models
- Progress assessments
- Expert Interaction
- Final presentation

PROGRAMME OUTCOMES

- PO1** Build end-to-end data science solutions from real datasets.
- PO2** Apply ML algorithms with validation for healthcare applications.
- PO3** Design CNN-based architectures & transfer learning pipelines.
- PO4** Analyze complex data using graph & network analytics.
- PO5** Present validated solutions through collaborative projects.

Programme Duration: June 8, 2026 - June 26, 2026

- 13 days of learning, 2 days of capstone project
- 10 days of online and 5 days of offline classes
- 3 hour theory + 2 hours hands -on per day
- 5 days a week

Centre coordinator:

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FEE INCLUDES (₹6,000 / participant)

- Welcome kit
- Food during offline sessions
- Accommodation on offline days
- Hands-on training

Registration: Scan here

